Psychopharmacology (2013) 225:627–635
DOI 10.1007/s00213-012-2850-6

ORIGINAL INVESTIGATION

Comparison of metabolic effects of ziprasidone versus olanzapine treatment in patients with first-episode schizophrenia

Jian-Jun Ou · Yi Xu · Hong-Hui Chen · Xiaoduo Fan · Keming Gao · Juan Wang · Xiao-Feng Guo · Ren-Rong Wu · Jing-Ping Zhao

Received: 9 June 2012 / Accepted: 10 August 2012 / Published online: 28 August 2012
© Springer-Verlag 2012

Abstract
Objective The objective of the study was to compare metabolic effects of ziprasidone versus olanzapine treatment in patients with first-episode schizophrenia.
Methods In this 6-week, multicenter, open-label trial, 260 patients were randomly assigned to receive ziprasidone or olanzapine treatment (130 per group). Primary metabolic measures were changes in weight and body mass index (BMI). Secondary metabolic measures were changes in glucose, insulin, lipids, and blood pressure. Efficacy and safety were also measured additionally.
Results A total number of 230 patients completed the study. The mean daily dosages were 138.2(28.6) mg for ziprasidone and 19.0(2.3) mg for olanzapine. After 6-week treatment, there were significant between-group differences in change scores on weight [4.22(3.49) kg versus −0.84(2.04) kg, $p < 0.001$] and BMI [1.59(1.37) versus −0.30(0.74), $p < 0.001$]. In addition, there were significant between-group differences in change scores on fasting plasma glucose, insulin, homeostasis model assessment 2-insulin resistance, low-density lipoprotein, total cholesterol, and triglycerides ($p < 0.001$); all the changes were clinically in favor of ziprasidone treatment. Both medications were effective in improving schizophrenia symptoms, but the decreases in Positive and Negative Syndrome Scale total scores of the olanzapine group were significantly greater than that of the ziprasidone group ($p < 0.05$). Compared with olanzapine, ziprasidone also induced more prolonging of corrected QT interval and extrapyramidal side effects ($p < 0.05$). Both medications were well tolerated, and no serious adverse events were observed in either group.
Conclusions Compared with olanzapine, ziprasidone treatment was associated with less adverse effects on glucose and lipid metabolism in patients with first-episode schizophrenia.

Keywords Ziprasidone · Olanzapine · First-episode schizophrenia · Metabolic effect

J.-J. Ou · J. Wang · X.-F. Guo · R.-R. Wu (✉) · J.-P. Zhao (✉)
Institute of Mental Health of the Second Xiangya Hospital, Central South University,
139 Middle Renmin Road,
Changsha 410011 Hunan, People's Republic of China
e-mail: wurenrong2005@yahoo.com.cn
e-mail: zhaojingpinghunan@163.com

Y. Xu
The First Affiliated Hospital of Medical School of Zhejiang University,
79 Qingchun Road,
Hangzhou 310003 Zhejiang, People's Republic of China

H.-H. Chen
Wuhan Mental Health Center,
70 Qiaokou Youyi Road,
Wuhan 430022 Hubei, People's Republic of China

X. Fan
UMass Memorial Medical Center,
University of Massachusetts Medical School,
Biotech One, Suite 100, 365 Plantation Street,
Worcester, MA 01605, USA

K. Gao
Department Psychiatry, Case Western Reserve/University Hospital Case Medical Center,
Cleveland, OH 44106, USA

J.-P. Zhao
The Hangzhou Seventh People Hospital,
Hangzhou 310001 Zhejiang, People's Republic of China

Introduction

Atypical antipsychotics have been widely used as the first-line treatment in schizophrenia (Leucht et al. 2003). However, atypical antipsychotics are associated with an increased risk of metabolic changes, such as weight gain and abnormalities in glucose metabolism and lipids, which

affect treatment adherence and also contribute to medical morbidity and mortality in this patient population (McEvoy et al. 2005; Miller et al. 2006). Previously, studies have suggested that people with severe mental illness can die up to 2 decades earlier than the general population (Tiihonen et al. 2009); heart disease, which is associated with metabolic disorders, is a leading cause of death in these patients (Mottillo et al. 2010).

Atypical antipsychotics vary in their propensity to induce metabolic changes (Henderson 2007). Some studies have suggested that olanzapine causes more metabolic problems than ziprasidone (Komossa et al. 2009, 2010). Switching from olanzapine to ziprasidone can improve metabolic conditions including weight, glucose metabolism, and lipids (Garman et al. 2007; Alptekin et al. 2009). Further, several studies have directly compared metabolic effects between olanzapine and ziprasidone treatment. In a 6-week double-blind study (N = 269) and the subsequent 6-month double-blind continuation study (n = 126), Simpson et al (2004; 2005) found a more favorable metabolic profile with ziprasidone treatment compared to olanzapine treatment, including less weight gain and lower fasting insulin, triglycerides, and cholesterol. In another 28-week double-blind study (n = 548), Breier et al (2005) found that, compared to olanzapine, ziprasidone was associated with a better metabolic profile in fasting blood glucose, high-density lipoprotein (HDL) and low-density lipoprotein (LDL). Moreover, In the Clinical Antipsychotic Trials of Intervention Effectiveness study (n = 1,493), olanzapine caused more metabolic disturbances than other antipsychotics including ziprasidone (Lieberman et al. 2005).

A few studies have compared metabolic effects of olanzapine versus ziprasidone treatment in recent-onset schizophrenia. In an 8-week double-blind study (n = 74), Grootens et al (2011) reported that, in patients with recent-onset schizophrenia or schizoaffective disorder, olanzapine caused more weight gain than ziprasidone; ziprasidone was associated with decreased levels of triglycerides and total cholesterols, while these parameters increased in the olanzapine group. However, the study did not assess levels of insulin, HDL, and LDL, and had a relatively small sample size. In the European First-Episode Schizophrenia Trial study (n = 498), there was more weight gain in the olanzapine group than in the ziprasidone group, but no significant differences were found between the two groups in glucose and lipid parameters. However, it is noticeable that most patients enrolled in the study already developed various metabolic problems at baseline (Kahn et al. 2008). We now report the results from a 6-week study comparing metabolic effects of olanzapine versus ziprasidone treatment in patients with first-episode schizophrenia.

Methods

Subjects

Male or female patients within the age range of 18–45 years old were recruited from The Second Xiangya Hospital of Central South University, The First Affiliated Hospital of Medical School of Zhejiang University, and Wuhan Mental Health Center between March 2009 and November 2010. The diagnosis of schizophrenia was determined by a research psychiatrist using the Diagnostic and Statistical Manual of Mental Disorders, fourth edition, text revision. Inpatient or outpatient patients were included if they meet the following conditions: the duration of illness was <1 year, which was evaluated depending on the history of present illness that was reported by the patients and guardians; the duration of antipsychotic treatment was less than two weeks before screening; the Positive and Negative Syndrome Scale (PANSS) total score was ≥60; the Clinical Global Impression-severity scale (CGI-S) score was ≥4 (at least "moderately ill").

Patients were excluded if they were in a special diet or special physical exercise program for weight loss. Female patients who were planning to be pregnant, or during pregnancy or lactation, were excluded from the study. Patients with certain medical conditions including heart disease, epilepsy, hepatic or renal diseases, diabetes, aplastic anemia, systemic lupus erythematosus, or asthma were excluded as well. Additionally, patients with a psychiatric diagnosis other than schizophrenia (determined by a research psychiatrist) were also excluded from the study.

Study design

The study was a multicenter, 6-week, open-label, parallel-group, randomized, trial. The study protocol was approved by the ethics committee of the Second Xiangya Hospital and was carried out in accordance with the Guideline for Good Clinical Practice and the Declaration of Helsinki. All patients or their legal guardian provided written informed consent before enrollment.

After screening, the current antipsychotic medication was tapered off within 10 days. Meanwhile, patients were randomized to one of the two treatment groups (olanzapine or ziprasidone) in the ratio of 1:1. They received a fixed dose of either ziprasidone 40 mg/day or olanzapine 5 mg/day for the first 3 days. From day 4 onward, the dose regimen was flexible, but higher than the initial dose. The maximum dose was 160 mg/day for ziprasidone or 20 mg/day for olanzapine. Once the study medication was titrated to an optimal dose based on patient's clinical response and tolerability, the same dose would be maintained until the end of study.

Certain psychotropic medications were allowed during the study: alprazolam for insomnia; trihexyphenidyl hydrochloride for dyskinesia, parkinsonism, or dystonia; and propranolol for akathisia or tachycardia. However, no other antipsychotics besides study medications were allowed during the study.

The sample size was calculated according to the Drug Registration Regulation of The People's Republic of China, which required 100 patients per group in order to achieve sufficient power (0.80). Assuming a 20–25 % attrition rate, 30 additional patients per group were included to ensure sufficient power. Therefore, each group consisted of 130 patients.

Metabolic parameters, efficacy, and safety were assessed at baseline, week 2, week 4, and week 6 (or endpoint).

Assessment of metabolic effects

Primary metabolic measures were changes of body weight and body mass index (BMI) from baseline to endpoint. Body weight was checked at baseline, week 2, week 4, and week 6 (or endpoint). The BMI was calculated as weight in kilograms divided by height in meters squared. In addition, the proportions of patients with significant weight gain were analyzed. Significant weight gain was defined as weight gain more than 7 % (Sachs and Guille 1999).

Secondary metabolic measures were changes of fasting plasma glucose (FPG), fasting insulin, homeostasis model assessment 2-β-cell function (HOMA2-%B), homeostasis model assessment 2-insulin sensitivity (HOMA2-%S), homeostasis model assessment 2-insulin resistance (HOMA2-IR), LDL, HDL, total cholesterol, triglycerides, and blood pressure from baseline to endpoint. These parameters were assessed at baseline and week 6 (or endpoint). HOMA2-%B, HOMA2-%S, and HOMA2-IR were calculated by HOMA2 Calculator (Wallace et al. 2004) (http://www.dtu.ox.ac.uk/homacalculator/index.php). In addition, the percentage of patients in the two groups that met the WHO criteria for metabolic syndrome was counted at baseline and week 6.

Efficacy and safety measures

The PANSS and CGI-S were administered at baseline and week 6 (or endpoint). One investigator per treatment center had completed the rater training and especially responsible for evaluating all the scales; the inter-rater reliability was good (Cronbach's $\alpha = 0.87$). All adverse events, regardless of the causal relationship with the study medications, were assessed. Vital signs were performed at baseline, week 2, week 4, and week 6 (or endpoint). All patients underwent electrocardiogram evaluation by a cardiologist at baseline and at week 6 (or endpoint). Laboratory safety measures were also performed at baseline and week 6 (or endpoint), including routine blood and urine tests, urine pregnancy test (female patients), liver and renal function tests, and hepatitis virus test.

Discontinuation

Most patients were accompanied by their caregivers during each visit. Medication treatment was supervised by inpatient nurses or caregivers at home. The participant's compliance with ziprasidone or olanzapine treatment for each visit interval was defined as taking more than 80 % of the study drug dosage prescribed for that interval. Those who were unable to demonstrate adequate compliance were withdrawn from the study. In addition, patients might be withdrawn from the study for safety concerns under the discretion of investigators.

Statistical analysis

Data analysis was based on the intent-to-treat population with last observation carried forward. Continuous variables were described using mean (standard deviation, SD), whereas categorical variables were reported using frequencies and percentages. Baseline between-group comparisons were performed using *t* test for continuous variables and χ^2 test or Fisher's exact test for categorical variables. The between-group comparison on treatment discontinuation was performed using Kaplan–Meier analysis.

Within-group comparisons in changes in continuous variables across different time points were examined using repeated measures analysis of variance (ANOVA). Between-group comparisons in changes from baseline to week 6 (or endpoint) in continuous variables were performed using analysis of covariance (ANCOVA) with treatment and treatment center as factors, and baseline values as covariate. In addition, χ^2 test or Fisher's exact test was used to compare between-group differences in percentages of patients with clinically significant metabolic conditions and adverse events. Further, logistic regression was used to examine whether age, sex, and baseline BMI could predict >7 % weight gain in week 6 (or endpoint).

All statistical tests were two tailed with significance level defined as alpha = 5 %. SPSS 19.0 (SPSS Inc, Chicago, IL) was used for statistical analysis.

Results

Characteristics of patients

A total number of 317 patients were screened for eligibility, of whom 260 were randomized to either the ziprasidone arm ($n = 130$) or the olanzapine arm ($n = 130$); 230 patients completed the study (Fig. 1). During the 6-week

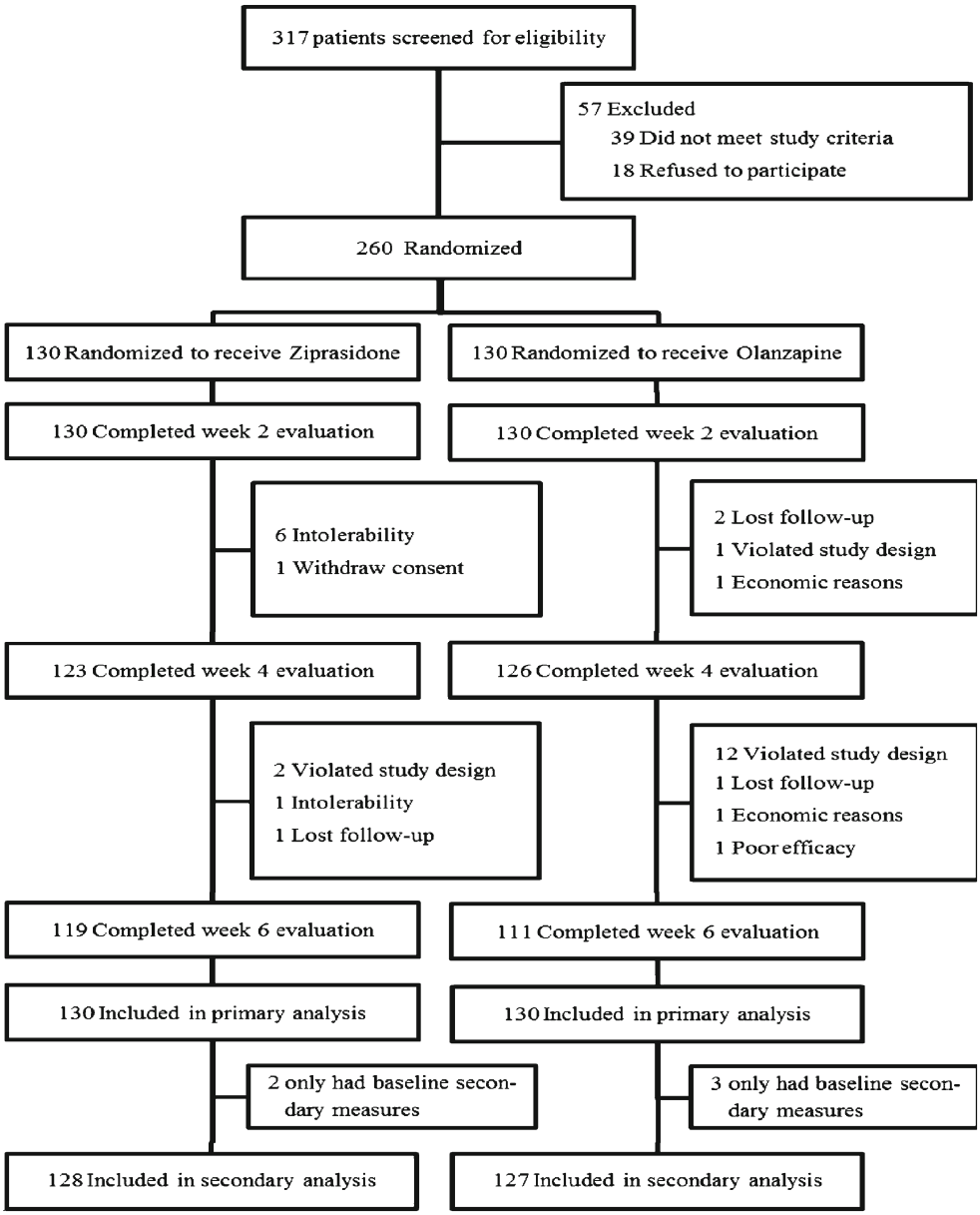

Fig. 1 Flowchart of study participants

Table 1 Baseline characteristics of the study sample ($N = 260$)

	Ziprasidone ($N = 130$)	Olanzapine ($N = 130$)	Test Statistics	p value
Age, mean(SD), years	26.86 (7.75)	27.68 (8.04)	−0.841	0.401
Gender, no (%)			0.389	0.533
Male	69(53.1)	74(56.9)		
Female	61(46.9)	56(43.1)		
Duration of illness, mean (SD), months	7.95(3.57)	8.43(4.41)	−0.958	0.339
Weight, mean (SD), kg	57.98(11.27)	56.54(10.37)	1.068	0.286
Body mass index, mean (SD)	21.17(3.50)	20.69(2.81)	1.209	0.228
PANSS total score, mean (SD)	103.87(17.48)	101.98(13.04)	0.989	0.323
CGI-S total score, mean (SD)	5.69(0.65)	5.65(0.63)	0.486	0.628
Corrected QT interval, mean (SD), ms	400.79(25.66)	396.78(22.82)	1.004	0.317

Continuous variables were analyzed by t test, and gender was analyzed by χ^2 test

trial, 11 (8.5 %) patients in the ziprasidone group and 19 (14.6 %) in the olanzapine arm discontinued the study; there was no significant difference in discontinuation rates between the two groups (p = 0.120). The Kaplan–Meier analysis for treatment discontinuation in time revealed no significant difference between the two groups (χ^2 = 2.142; df = 1; p = 0.143). The mean daily dosages of study medications were 138.2 (28.6) mg for ziprasidone and 19.0 (2.3) mg for olanzapine (values expressed as mean (SD)); the median dosages of study medications were 160 mg for ziprasidone and 20 mg for olanzapine. There were no significant differences in baseline demographic and clinical characteristics between the two groups (Table 1).

Metabolic effects

Within the olanzapine group, there was significant weight gain at week 2 ($p < 0.05$), and weeks 4 and 6 ($p < 0.001$) compared to baseline values; similarly, there was significant increase in BMI at weeks 2, 4, and 6 ($p < 0.001$). Within the ziprasidone group, there was significant weight loss at week 2 ($p < 0.05$), and weeks 4 and 6 ($p < 0.001$); there was significant decrease in BMI at week 2 ($p < 0.05$) and weeks 4 and 6 ($p < 0.001$) (Table 2).

There were significant between-group differences in weight change and BMI change controlling for baseline values at weeks 2, 4, and 6 (Fig. 2). At week 6, ziprasidone-treated patients had a significant weight loss of 0.84 (2.04) kg, whereas olanzapine-treated patients had a weight gain of 4.22 (3.49) kg ($p < 0.001$). Similarly, the BMI decreased by 0.30 (0.74) in the ziprasidone group while it increased by 1.59 (1.37) in the olanzapine group ($p < 0.001$).

Additional analyses showed that 10.0 % (13/130), 30.0 % (39/130), and 43.8 % (57/130) olanzapine-treated patients had a weight gain of more than 7 % at weeks 2, 4, and 6, respectively. In contrast, no one in the ziprasidone group showed a weight gain of more than 7 % at each time point (Fisher's exact test, $p < 0.05$). Logistic regression did not seem to suggest that age, sex, and baseline BMI can predict >7 % weight gain in olanzapine-treated patients (age, p = 0.769; sex, p = 0.287; baseline BMI, p = 0.214).

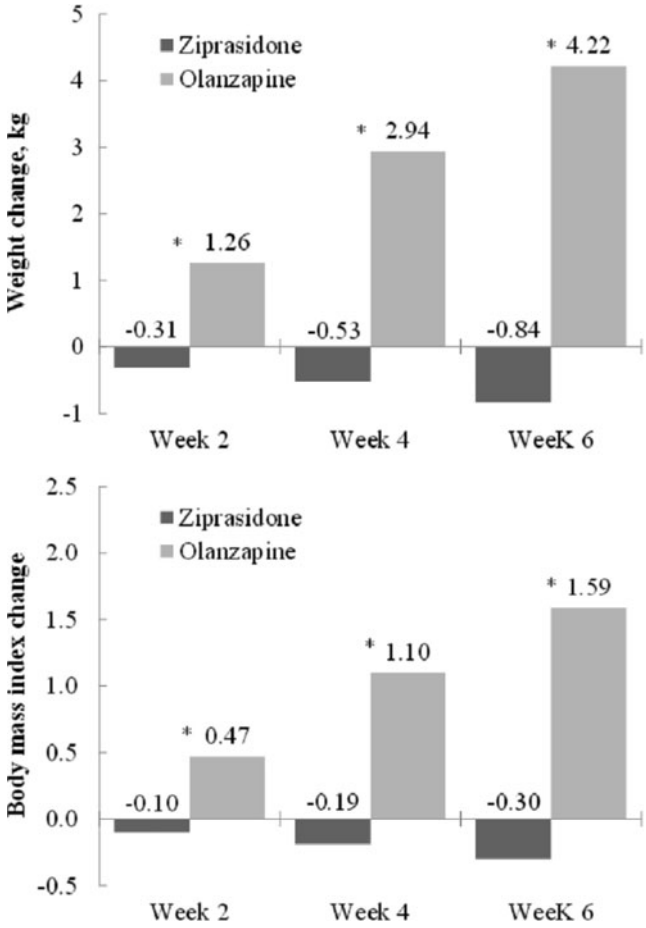

Fig. 2 Changes in weight and body mass index during the 6-week treatment with ziprasidone or olanzapine. The *asterisks* indicate significant difference between the 2 groups controlling for baseline values, $p < 0.001$

Table 2 Weight and body mass index during the 6-week treatment with ziprasidone or olanzapine

	Time point	Ziprasidone mean (SD) (N = 130)	Olanzapine mean (SD) (N = 130)	Between-Group difference	
				F	p value
Weight, kg	Week 2	57.67(11.21)*	57.80(10.23)**	79.318	<0.001
	Week 4	57.44(10.92) **	59.48(10.35) **	161.560	<0.001
	Week 6	57.14(10.72) **	60.77(10.45) **	204.730	<0.001
Body mass index	Week 2	21.06(3.52) *	21.16(2.74) **	73.634	<0.001
	Week 4	20.97(3.38) **	21.79(2.82) **	155.280	<0.001
	Week 6	20.87(3.34) **	22.28(2.93) **	189.796	<0.001

Data analyses were based on last observation carried forward. Compared with baseline, within-group differences were analyzed using repeated ANOVA. Between-group differences at weeks 2, 4, and 6 were analyzed using ANCOVA controlling for baseline values

* $p < 0.05$; ** $p < 0.001$

Within the olanzapine group, there were significant increases in FPG, insulin, HOMA2-%S, HOMA2-IR, LDL, HDL, total cholesterol, and triglycerides at week 6 ($p < 0.05$); there was a significant decrease in HOMA2-%S at week 6 ($p < 0.001$). Within the ziprasidone group, there were significant increases in HOMA2-%B, HDL, and triglycerides at week 6 ($p < 0.05$); there was a significant decrease in FPG at week 6 ($p < 0.05$) (Table 3).

Further analyses showed significant between-group differences in changes of various metabolic parameters including FPG, insulin, HOMA2-%S, HOMA2-IR, LDL, total cholesterol, and triglycerides at week 6 after controlling for baseline values ($p < 0.001$). There were no significant differences in changes of HOMA2-%B, HDL, and systolic and diastolic blood pressures ($p > 0.2$) (Table 3). There was not any difference for the percentage of patients in the two groups that met the criteria for metabolic syndrome (WHO 1999) at baseline [ziprasidone 3(2.3 %) versus olanzapine 1 (0.8 %), $p = 0.622$] and week 6 [ziprasidone 3(2.3 %) versus olanzapine 7 (5.5 %), $p = 0.216$].

Table 3 Changes in metabolic parameters

	Baseline, mean(SD)	Week 6, mean(SD)	Change score, mean(SD)	Between-group difference in change score	
				F	*p* value
Fasting Plasma Glucose, mmol/L					
Ziprasidone (*N* = 128)	4.55 (0.50)	4.40 (0.50)*	−0.15 (0.63)	78.977	<0.001
Olanzapine (*N* = 127)	4.58 (0.41)	4.94 (0.50)**	0.37 (0.59)		
Insulin, mU/L					
Ziprasidone (*N* = 128)	8.43 (5.19)	9.15 (7.47)	0.72 (5.34)	25.412	<0.001
Olanzapine (*N* = 127)	9.54 (5.64)	15.44 (12.45)**	5.90 (9.84)		
HOMA2-%B, %					
Ziprasidone (*N* = 128)	123.62 (44.61)	135.17 (65.77)*	15.57 (49.48)	0.206	0.650
Olanzapine (*N* = 127)	126.07 (48.33)	147.38 (71.33)*	17.28 (60.28)		
HOMA2-%S, %					
Ziprasidone (*N* = 128)	116.23 (59.45)	129.94 (73.92)	3.71 (61.93)	42.916	<0.001
Olanzapine (*N* = 127)	113.39 (55.02)	79.54 (50.09)**	−29.41 (33.76)		
HOMA2-IR,					
Ziprasidone (*N* = 128)	1.12 (0.64)	1.13 (0.90)	0.06 (0.65)	31.401	<0.001
Olanzapine (*N* = 127)	1.11 (0.57)	1.97 (1.51)**	0.75 (1.16)		
Low-density lipoprotein , mmol/L					
Ziprasidone (*N* = 128)	2.11 (0.65)	2.14 (0.51)	0.02 (0.49)	16.318	<0.001
Olanzapine (*N* = 127)	2.16 (0.43)	2.39 (0.55)**	0.25 (0.46)		
High-density lipoprotein, mmol/L					
Ziprasidone (*N* = 128)	1.36 (0.31)	1.41 (0.29)*	0.06 (0.22)	1.326	0.251
Olanzapine (*N* = 127)	1.32 (0.40)	1.43 (0.41)*	0.11 (0.35)		
Total cholesterol, mmol/L					
Ziprasidone (*N* = 128)	3.97 (0.69)	4.06 (0.74)	0.09 (0.65)	22.389	<0.001
Olanzapine (*N* = 127)	4.04 (0.68)	4.51 (0.80)**	0.47 (0.94)		
Triglycerides, mmol/L					
Ziprasidone (*N* = 128)	1.18 (0.46)	1.30 (0.49)*	0.12 (0.43)	13.175	<0.001
Olanzapine (*N* = 127)	1.19 (0.54)	1.56 (0.72)**	0.36 (0.73)		
Systolic blood pressure, mm Hg					
Ziprasidone (*N* = 128)	117.66 (13.61)	117.43 (13.13)	−0.23 (9.79)	0.198	0.657
Olanzapine (*N* = 127)	117.68 (10.64)	117.97 (10.50)	0.34 (7.38)		
Diastolic blood pressure, mm Hg					
Ziprasidone (*N* = 128)	75.81 (8.06)	75.93 (7.90)	0.11 (8.17)	0.021	0.884
Olanzapine (*N* = 127)	76.88 (6.71)	76.30 (5.65)	−0.51 (5.29)		

Data analyses were based on last observation carried forward. Within-group differences were analyzed using repeated ANOVA. Between-group differences in change scores at week 6 were analyzed using ANCOVA controlling for baseline values

* $p < 0.05$; ** $p < 0.001$

Efficacy and safety

As shown in Table 4, there was a significant decrease in the PANSS total score within each group at week 6 ($p < 0.001$); there was a significant between-group difference in the PANSS total score and corrected QT interval changes after controlling for baseline values at week 6 ($p < 0.05$). However, there was no significant between-group difference in the CGI-S score change. More patients in ziprasidone group had extrapyramidal side effects than in olanzapine group ($p = 0.001$).

Although ten patients had mild elevations in liver transaminases at week 6 or study endpoint, no significant difference was found between the two groups. No patient in either group exhibited a corrected QT interval ≥ 500 ms or other clinical significant abnormalities in electrocardiogram, and no serious adverse events were observed in either group.

Discussion

This multicenter, randomized, and open-label study compared the metabolic effects of ziprasidone and olanzapine in patients with first-episode schizophrenia who had duration of illness less than 1 year.

The weight gain findings in olanzapine-treated patients are consistent with previous studies (Simpson et al. 2004; Brown and Estoup 2005; Grootens et al. 2011). In an 8-week study, Grootens et al (2011) found that a mean dose of 14 mg olanzapine was associated with a mean weight gain of 6.8 kg in patients with recent-onset schizophrenia. However, the mean weight gain of 4.22 kg in our study is relatively higher than in other short-term studies in chronic patients with schizophrenia (Simpson et al. 2004; Dossenbach et al. 2007; Lublin et al. 2009). The weight loss findings in ziprasidone-treated patients in our study are consistent with some but not all previous studies (Simpson et al. 2004; Breier et al. 2005; Brown and Estoup 2005). The differences in weight response, for both olanzapine and ziprasidone, across different studies might be explained by various factors, such as duration of illness, previous exposure to other antipsychotic medications, duration of study intervention, and dosage of study medication.

Our study also demonstrated that olanzapine was associated with significant increases in FPG, insulin, and HOMA2-IR, whereas ziprasidone had less effect on glucose metabolism. The results were consistent with some previous studies (Simpson et al. 2004; Rummel-Kluge et al. 2010). HOMA2-%B and HOMA2-%S reflect β-cell function and insulin sensitivity, respectively (Wallace et al. 2004). In our study, a short-term 6-week olanzapine treatment was associated with increased HOMA2-%B and decreased HOMA2-%S, whereas ziprasidone had less impact on these two parameters. Our results suggest that ziprasidone could be considered as an alternative to improve other antipsychotic-induced hyperglycemia and insulin resistance (Brown and Estoup 2005).

Elevated levels of LDL and total cholesterol are both established risk factors for cardiovascular disease (Law and Wald 1994; Ip et al. 2009; Olsson 2010). Consistent with most previous findings (Simpson et al. 2004; Breier et al. 2005; Brown and Estoup 2005; Grootens et al. 2011), our results demonstrated that olanzapine caused more elevations in both LDL and total cholesterol than ziprasidone. One previous study did not find significant difference on lipid changes between ziprasidone and olanzapine treatment (Kahn et al. 2008), which might be explained by the high rate of lipid abnormalities at baseline. Elevated HDL is considered a protective factor for cardiovascular disease (Gordon et al. 1989), whereas elevated triglycerides is a risk factor for it, independent of HDL cholesterol (Hokanson and Austin 1996). In our study, elevated HDL was found in both groups; elevated triglycerides was also found in both groups, but more so in the olanzapine group. Results from similar previous studies

Table 4 Comparison of efficacy and safety between Ziprasidone and Olanzapine groups

	Ziprasidone (N = 130)	Olanzapine (N = 130)	Test statistics	p
Week 6 PANSS total score, mean (SD)	60.48 (18.12)**	55.65 (16.39)**	4.078	0.044
Change of PANSS total score, mean (SD)	−43.39 (18.26)	−46.32 (18.41)	4.078	0.044
Week 6 CGI-S total score, mean (SD)	2.93 (1.16)**	2.98 (1.05)**	0.203	0.653
Change of CGI-S total score, mean (SD)	−2.76 (1.19)	−2.68 (1.11)	0.203	0.653
Week 6 corrected QT interval, mean (SD), ms	417.72 (30.92)**	402.96 (24.24)*	9.785	0.002
Change of corrected QT interval, mean (SD), ms	16.93 (29.71)	6.18 (23.37)	9.785	0.002
Extrapyramidal side effects, N (%)	51 (39.23)	26 (20.00)	11.532	0.001
Increased transaminase, N (%)	3 (2.31)	7 (5.38)		0.334

Compared with baseline, within-group differences were analyzed using repeated ANOVA. Between-group differences of continuous data were analyzed by ANCOVA controlling for baseline values, and qualitative data were analyzed by χ^2 or Fisher's exact test. Adverse events >3 % are shown, and no serious adverse events were observed in both groups

* $p < 0.05$, ** $p < 0.001$

are inconsistent on changes in HDL and triglycerides, which again might be explained by different lipid levels at baseline.

In our study, clinical symptoms were significantly improved in both groups with a relative superiority in the olanzapine group. The decreases in PANSS total scores of the olanzapine group were significantly greater than that of the ziprasidone group. The difference of the two groups in mean changes of PANSS total scores from baseline to week 6 was only 3, so the clinical significance was not clear. Ziprasidone was associated with more extrapyramidal side effects than olanzapine in our study; this finding is consistent with some previous studies and recent Cochrane data (Breier et al. 2005; Komossa et al. 2009, 2010). Although we did not collect information on the use of anti-Parkinson medications in our study, some previous studies have found that patients with ziprasidone treatment used more anti-Parkinson medications than with olanzapine treatment (Grootens et al. 2011; Rummel-Kluge et al. 2012). More elevations in corrected QT interval were observed in ziprasidone-treated patients than in olanzapine-treated patients; however, the clinical implication of this finding is uncertain. A recent large sample study ($N = 18{,}154$) demonstrated that ziprasidone is not associated with an elevated risk of non-suicidal mortality relative to olanzapine in real-world use (Strom et al. 2011). In addition, our study found that some patients had mild elevations in liver transaminases during the 6-week treatment, which suggests that it is necessary to monitor liver function during olanzapine or ziprasidone treatment.

In conclusion, ziprasidone and olanzapine had different impact on body weight, glucose metabolism, and lipids in patients with first-episode schizophrenia; the changes in these metabolic parameters were clinically in favor of ziprasidone treatment. Unfavorable metabolic changes with olanzapine treatment could be observed during the early phase of treatment. Both medications were effective in improving clinical symptom, and well tolerated. However, compared with ziprasidone, olanzapine was better in decreasing PANSS total score, and induced less prolonging of corrected QT interval and extrapyramidal side effects.

Study limitations

There were several limitations in our study. First, the open-label trial design might introduce biases from both patients and study investigators; second, the study time period was relatively short. Third, even though patients were asked about history of substance use during the screening process, urine test was not done to confirm the status of substance use. Schizophrenia is a chronic, devastating, mental illness, for which many patients will need to be on antipsychotic treatment in their lifetime. Future studies to evaluate and compare long term metabolic impact of olanzapine and ziprasidone treatment in first-episode schizophrenia are needed.

Acknowledgments This study was supported by the Pfizer Inc., the National Natural Science Foundation of China (Grant No. 30971052 to RRW), and National R&D Special Program for Health Professions (Grant No. 201002003 to JPZ). Dr. Wu expresses her gratitude to the World Psychiatric Association for providing financial support for her fellowship.

Conflict of interest All authors declared no conflict of interest. The only role of Pfizer Inc. was in the monitoring. The company had no further role in the study design, collection, analysis, and interpretation of data and writing the reports. The company also had no role in the decision to submit the paper for publication.

References

Alptekin K, Hafez J, Brook S, Akkaya C, Tzebelikos E, Ucok A, El Tallawy H, Danaci AE, Lowe W, Karayal ON (2009) Efficacy and tolerability of switching to ziprasidone from olanzapine, risperidone or haloperidol: an international, multicenter study. Int Clin Psychopharmacol 24:229–238

Breier A, Berg PH, Thakore JH, Naber D, Gattaz WF, Cavazzoni P, Walker DJ, Roychowdhury SM, Kane JM (2005) Olanzapine versus ziprasidone: results of a 28-week double-blind study in patients with schizophrenia. Am J Psychiatry 162:1879–1887

Brown RR, Estoup MW (2005) Comparison of the metabolic effects observed in patients treated with ziprasidone versus olanzapine. Int Clin Psychopharmacol 20:105–112

Dossenbach M, Treuer T, Kryzhanovskaya L, Saylan M, Dominguez S, Huang X (2007) Olanzapine versus chlorpromazine in the treatment of schizophrenia: a pooled analysis of four 6-week, randomized, open-label studies in the Middle East and North Africa. J Clin Psychopharmacol 27:329–337

Garman PM, Ried LD, Bengtson MA, Hsu C, McConkey JR (2007) Effect on lipid profiles of switching from olanzapine to another second-generation antipsychotic agent in veterans with schizophrenia. J Am Pharm Assoc: JAPhA 47:373–378

Gordon DJ, Probstfield JL, Garrison RJ, Neaton JD, Castelli WP, Knoke JD, Jacobs DR Jr, Bangdiwala S, Tyroler HA (1989) High-density lipoprotein cholesterol and cardiovascular disease. Four prospective American studies. Circulation 79:8–15

Grootens KP, van Veelen NM, Peuskens J, Sabbe BG, Thys E, Buitelaar JK, Verkes RJ, Kahn RS (2011) Ziprasidone vs olanzapine in recent-onset schizophrenia and schizoaffective disorder: results of an 8-week double-blind randomized controlled trial. Schizophr Bull 37:352–361

Henderson DC (2007) Weight gain with atypical antipsychotics: evidence and insights. J clin psychiatry 68(Suppl 12):18–26

Hokanson JE, Austin MA (1996) Plasma triglyceride level is a risk factor for cardiovascular disease independent of high-density lipoprotein cholesterol level: a meta-analysis of population-based prospective studies. J Cardiovasc Risk 3:213–219

Ip S, Lichtenstein AH, Chung M, Lau J, Balk EM (2009) Systematic review: association of low-density lipoprotein subfractions with cardiovascular outcomes. Ann Intern Med 150:474–484

Kahn RS, Fleischhacker WW, Boter H, Davidson M, Vergouwe Y, Keet IP, Gheorghe MD, Rybakowski JK, Galderisi S, Libiger J, Hummer M, Dollfus S, Lopez-Ibor JJ, Hranov LG, Gaebel W, Peuskens J, Lindefors N, Riecher-Rossler A, Grobbee DE (2008) Effectiveness of antipsychotic drugs in first-episode schizophrenia and schizophreniform disorder: an open randomised clinical trial. Lancet 371:1085–1097

Komossa K, Rummel-Kluge C, Hunger H, Schmid F, Schwarz S, Duggan L, Kissling W, Leucht S (2010) Olanzapine versus other atypical antipsychotics for schizophrenia. Cochrane Database Syst Rev: CD006654

Komossa K, Rummel-Kluge C, Hunger H, Schwarz S, Bhoopathi PS, Kissling W, Leucht S (2009) Ziprasidone versus other atypical antipsychotics for schizophrenia. Cochrane Database Syst Rev: CD006627

Law MR, Wald NJ (1994) An ecological study of serum cholesterol and ischaemic heart disease between 1950 and 1990. Eur J Clin Nutr 48:305–325

Leucht S, Barnes TR, Kissling W, Engel RR, Correll C, Kane JM (2003) Relapse prevention in schizophrenia with new-generation antipsychotics: a systematic review and exploratory meta-analysis of randomized, controlled trials. Am J Psychiatry 160:1209–1222

Lieberman JA, Stroup TS, McEvoy JP, Swartz MS, Rosenheck RA, Perkins DO, Keefe RS, Davis SM, Davis CE, Lebowitz BD, Severe J, Hsiao JK (2005) Effectiveness of antipsychotic drugs in patients with chronic schizophrenia. N Engl J Med 353:1209–1223

Lublin H, Haug HJ, Koponen H, Sigmundsson T, Kolb SA (2009) Ziprasidone versus olanzapine, risperidone or quetiapine in patients with chronic schizophrenia: a 12-week open-label, multicentre clinical trial. world j biol psychiatry: off j World Fed Soc Biol Psychiatry 10:710–718

McEvoy JP, Meyer JM, Goff DC, Nasrallah HA, Davis SM, Sullivan L, Meltzer HY, Hsiao J, Scott Stroup T, Lieberman JA (2005) Prevalence of the metabolic syndrome in patients with schizophrenia: baseline results from the Clinical Antipsychotic Trials of Intervention Effectiveness (CATIE) schizophrenia trial and comparison with national estimates from NHANES III. Schizophr Res 80:19–32

Miller BJ, Paschall CB, Svendsen DP (2006) Mortality and medical comorbidity among patients with serious mental illness. Psychiatr Serv 57:1482–1487

Mottillo S, Filion KB, Genest J, Joseph L, Pilote L, Poirier P, Rinfret S, Schiffrin EL, Eisenberg MJ (2010) The metabolic syndrome and cardiovascular risk a systematic review and meta-analysis. J Am Coll Cardiol 56:1113–1132

Olsson AG (2010) HDL and LDL as therapeutic targets for cardiovascular disease prevention: the possible role of niacin. Nutr Metab Cardiovasc Dis 20:553–557

Rummel-Kluge C, Komossa K, Schwarz S, Hunger H, Schmid F, Kissling W, Davis JM, Leucht S (2012) Second-generation antipsychotic drugs and extrapyramidal side effects: a systematic review and meta-analysis of head-to-head comparisons. Schizophr Bull 38:167–177

Rummel-Kluge C, Komossa K, Schwarz S, Hunger H, Schmid F, Lobos CA, Kissling W, Davis JM, Leucht S (2010) Head-to-head comparisons of metabolic side effects of second generation antipsychotics in the treatment of schizophrenia: a systematic review and meta-analysis. Schizophr Res 123:225–233

Sachs GS, Guille C (1999) Weight gain associated with use of psychotropic medications. J clin psychiatry 60(Suppl 21):16–19

Simpson GM, Glick ID, Weiden PJ, Romano SJ, Siu CO (2004) Randomized, controlled, double-blind multicenter comparison of the efficacy and tolerability of ziprasidone and olanzapine in acutely ill inpatients with schizophrenia or schizoaffective disorder. Am J Psychiatry 161:1837–1847

Simpson GM, Weiden P, Pigott T, Murray S, Siu CO, Romano SJ (2005) Six-month, blinded, multicenter continuation study of ziprasidone versus olanzapine in schizophrenia. Am J Psychiatry 162:1535–1538

Strom BL, Eng SM, Faich G, Reynolds RF, D'Agostino RB, Ruskin J, Kane JM (2011) Comparative mortality associated with ziprasidone and olanzapine in real-world use among 18,154 patients with schizophrenia: the Ziprasidone Observational Study of Cardiac Outcomes (ZODIAC). Am J Psychiatry 168:193–201

Tiihonen J, Lonnqvist J, Wahlbeck K, Klaukka T, Niskanen L, Tanskanen A, Haukka J (2009) 11-year follow-up of mortality in patients with schizophrenia: a population-based cohort study (FIN11 study). Lancet 374:620–627

Wallace TM, Levy JC, Matthews DR (2004) Use and abuse of HOMA modeling. Diabetes Care 27:1487–1495